

高数B上. HW1.

No.

Date

习题 1.3.2.

证明: $\forall \varepsilon > 0. \exists N \in \mathbb{N}^+ \text{ s.t. } \forall n > N. \text{ 有 } |a_n - L| < \varepsilon$

从而, $\forall n > N. \text{ 有 } ||a_n| - |L|| \leq |a_n - L| < \varepsilon.$

故 $\lim_{n \rightarrow \infty} |a_n| = |L|. \quad \square.$

3. (1) 证明: $\forall \varepsilon > 0. \exists N \in \mathbb{N}^+ \text{ s.t. } \forall n > N. \text{ 有 } |a_n - L| < \varepsilon$

取 $\varepsilon = 1$, 即 $|a_n - L| < 1 \Rightarrow ||a_n| - |L|| < |a_n - L| < 1$
故 $|a_n| < |L| + 1. \quad \square.$

(2) 证明: 取 $M = \max \{|a_1|, |a_2|, \dots, |a_N|, |L| + 1\}$ 即可. $\square.$

4. (4) 由于 $|n!/n^n| \leq \frac{1}{n}$

故 $\forall \varepsilon > 0. \text{ 取 } N = [\frac{1}{\varepsilon}] + 1. \text{ 有 } \frac{1}{N} < \varepsilon.$

那么, $\forall n > N. \text{ 有 } |n!/n^n| \leq \frac{1}{n} < \frac{1}{N} < \varepsilon.$

从而 $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0. \quad \square.$

(5) 原式 = $1 - \frac{1}{n}$ (裂项). $|1 - \frac{1}{n} - 1| = \frac{1}{n}.$

故 $\forall \varepsilon > 0. \text{ 取 } N = [\frac{1}{\varepsilon}] + 1.$

有 $\forall n > N. \left| \left(\frac{1}{1 \times 2} + \dots + \frac{1}{(n-1)n} \right) - 1 \right| = \frac{1}{n} < \frac{1}{N} < \varepsilon.$

故 $\lim_{n \rightarrow \infty} \left[\frac{1}{1 \times 2} + \dots + \frac{1}{(n-1)n} \right] = 1. \quad \square.$

7. (1) $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1} + \sqrt{n}} = 0.$

(2) $\lim_{n \rightarrow \infty} \frac{n^3 + 3n^2 - 100}{4n^3 - n + 2} = \lim_{n \rightarrow \infty} \frac{1 + \frac{3}{n} - \frac{100}{n^3}}{4 - \frac{1}{n^2} + \frac{2}{n^3}} = \frac{1 + 0 - 0}{4 - 1 + 0} = \frac{1}{4}.$

(3) $\lim_{n \rightarrow \infty} \frac{(2n+4)^4}{n^4 + n^3} = \lim_{n \rightarrow \infty} \frac{(2 + \frac{4}{n})^4}{1 + \frac{1}{n}} = \frac{(2+0)^4}{1+0} = 16$

(4) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{-2n} = \left(\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n\right)^{-2} = e^{-2}.$

$$(1) \lim_{x \rightarrow a} \sqrt{x} = \sqrt{a} \quad (a > 0).$$

证明: $\forall \varepsilon > 0$. $|\sqrt{x} - \sqrt{a}| = \frac{|x-a|}{|\sqrt{x} + \sqrt{a}|}$

当 $|x-a| < \frac{3a}{4}$ 时, $\sqrt{x} > \sqrt{\frac{a}{4}} = \frac{1}{2}\sqrt{a}$. $\sqrt{x} + \sqrt{a} > \frac{3}{2}\sqrt{a}$.

只需对充分小的 δ 做分析, 因此可以自己给自己加合适的条件!
此时, $|\sqrt{x} - \sqrt{a}| < \frac{2}{3\sqrt{a}} |x-a|$.

故取 $\delta = \min\{\frac{3\sqrt{a}}{2}\varepsilon, \frac{3}{4}a\}$. 当 $|x-a| < \delta$ 时,

同时有 $|x-a| < \frac{3a}{4}$ 和 $|x-a| < \frac{3\sqrt{a}}{2}\varepsilon$.

因此, $|\sqrt{x} - \sqrt{a}| < \frac{2}{3\sqrt{a}} |x-a| < \frac{2}{3\sqrt{a}} \cdot \frac{3\sqrt{a}}{2}\varepsilon = \varepsilon$. $\square$

$$(3) \lim_{x \rightarrow a} e^x = e^a.$$

证明: $\forall \varepsilon > 0$. $|e^x - e^a| = e^a \cdot |e^{x-a} - 1|$
 $\leq e^a \cdot (e^{|x-a|} - 1)$
 因为对于 $t > 0$, $e^t - 1 = e^t(1 - e^{-t}) > 1 - e^{-t}$
 故 $|e^t - 1| > |e^{-t} - 1|$

分析: 接下来需要找 $\delta > 0$, s.t. $|x-a| < \delta$ 时,

$$e^a \cdot (e^{|x-a|} - 1) < e^a \cdot (e^\delta - 1) < \varepsilon.$$

这等同于 $e^\delta < e^{-a} \cdot \varepsilon + 1 \Leftrightarrow \delta < \ln(1 + e^{-a}\varepsilon)$

取 $\delta = \max\{\ln(e^{-a}\varepsilon), 0\} = \ln \varepsilon - a$. $\max\{\ln \varepsilon, 0\}$

那么, $|x-a| < \delta$ 时

取 $\delta = \frac{1}{2} \ln(1 + e^{-a}\varepsilon) < \ln(1 + e^{-a}\varepsilon)$. Rmk. 注意保证 $\delta > 0$.

那么, $|x-a| < \delta$ 时, $|e^x - e^a| \leq e^a \cdot (e^{|x-a|} - 1)$

$$< e^a \cdot (e^\delta - 1) < \varepsilon. \quad \square$$

$$3. (2) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x^2} = \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{2}}{x/2} \right)^2 = \frac{1}{2}.$$

$$(3) \lim_{x \rightarrow 0} \frac{\sqrt{x+a} - \sqrt{a}}{x} \quad (a > 0)$$

$$= \lim_{x \rightarrow 0} \frac{x}{(\sqrt{x+a} + \sqrt{a})x} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+a} + \sqrt{a}} = \frac{1}{2\sqrt{a}}.$$

$$(6) \lim_{x \rightarrow \infty} \frac{(2x-3)^{20} (3x+2)^{10}}{(2x+1)^{30}} = \frac{2^{20} \times 3^{10}}{2^{30}} = \left(\frac{3}{2}\right)^{10}.$$

$$(10) \lim_{x \rightarrow 1} \frac{x^n - 1}{x - 1} = \lim_{x \rightarrow 1} (x^{n-1} + x^{n-2} + \dots + 1) = n.$$

$$(11) \lim_{x \rightarrow \infty} (\sqrt{x^2+1} - \sqrt{x^2-1}) = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x^2+1} + \sqrt{x^2-1}} = 0.$$

$$(12) a_n / b_n$$

$$4. (1) \lim_{x \rightarrow 0} \frac{\sin \alpha x}{\tan \beta x} \quad (\beta \neq 0)$$

$$= \lim_{x \rightarrow 0} \frac{\sin \alpha x}{\alpha x} \cdot \lim_{x \rightarrow 0} \frac{\beta x}{\tan \beta x} \cdot \frac{\alpha}{\beta} = 1 \cdot 1 \cdot \frac{\alpha}{\beta} = \alpha / \beta.$$

$$(2) \lim_{x \rightarrow 0} \frac{\sin 2x^2}{3x} = \lim_{x \rightarrow 0} \frac{\sin 2x^2}{2x^2} \cdot \lim_{x \rightarrow 0} \frac{2x^2}{3x} = 1 \times 0 = 0.$$

$$(5) \lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a} = \lim_{x \rightarrow a} \frac{2 \sin \frac{x-a}{2} \cdot \cos \frac{x+a}{2}}{x-a}$$

$$= \lim_{x \rightarrow a} \frac{\sin \frac{x-a}{2}}{\frac{x-a}{2}} \cdot \lim_{x \rightarrow a} \cos \frac{x+a}{2} = 1 \cdot \cos a = \cos a$$

$$(6) \lim_{x \rightarrow \infty} \left(1 + \frac{k}{x}\right)^{-x} = \lim_{x \rightarrow \infty} \left(\left(1 + \frac{k}{x}\right)^{\frac{x}{k}}\right)^{-k}$$

$$= \left(\lim_{x \rightarrow \infty} \left(1 + \frac{k}{x}\right)^{\frac{x}{k}}\right)^{-k} = e^{-k}.$$